M.SC. MACHINE LEARNING · PROJECT DEFENSE

Energy Market Intelligence *System*

Forecasting & battery dispatch *under uncertainty* in the German–Luxembourg day-ahead electricity market.

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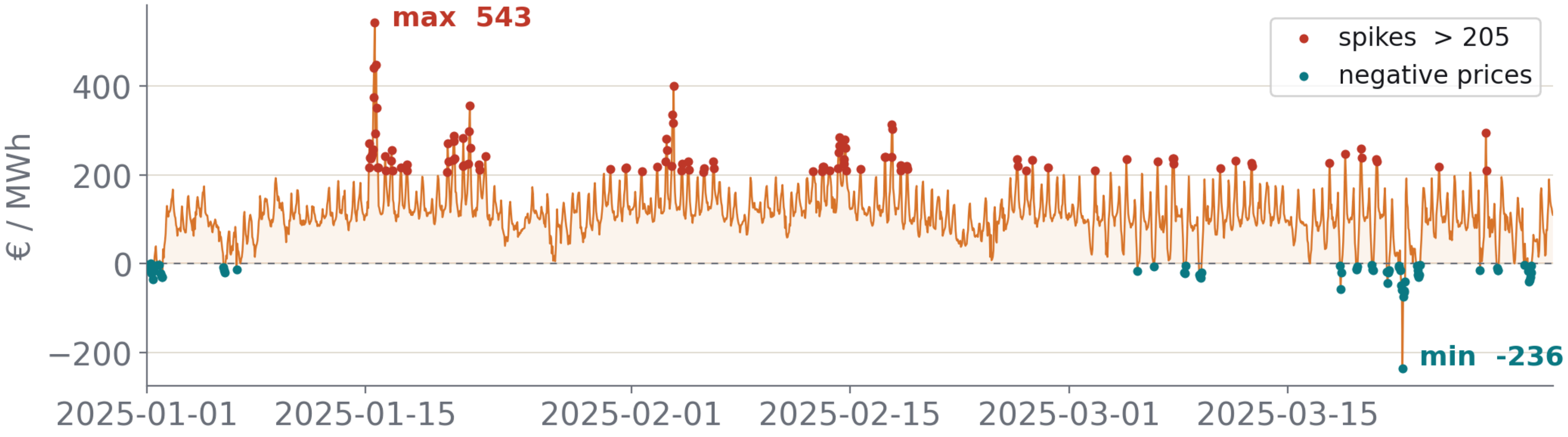
Free University of Bozen-Bolzano · June 2026

THE PROBLEM

Day-ahead prices defy point forecasts



Renewables, weather and geopolitics have made the market **unstable**: it no longer behaves the way it used to, so a single predicted number is fragile.

+€543

intraday spike, €/MWh

−€236

goes negative: surplus renewables, sellers pay to offload

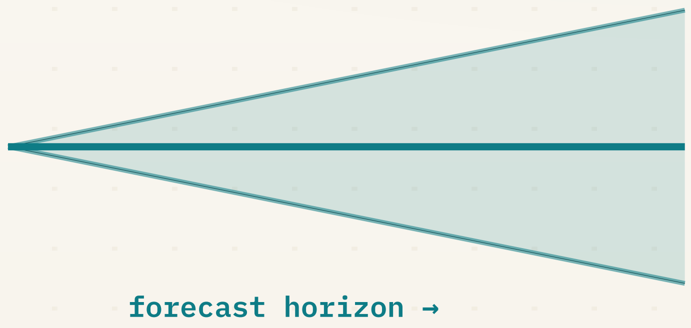
volatile

regime-shifting; the old patterns break

THE IDEA

Forecast the band, not the point

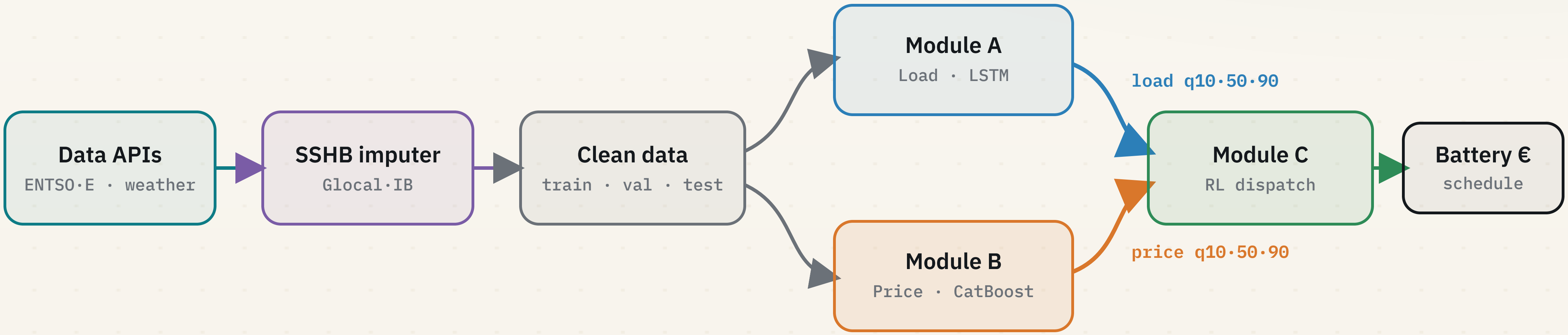
- We never predict **one number**. We predict a **band** (q10 / q50 / q90).
- We are confident the true value lands **inside** that band.
- The band is **propagated** downstream, then turned into **euros**.



Each stage admits what it doesn't know, and that propagated uncertainty is **what turns a battery into profit.**

SYSTEM ARCHITECTURE

One pipeline, five stages



Two forecasters and one decision agent, each consuming the previous stage's **quantiles**.

PART 01

01

Data & Imputation

Six messy external sources, repaired by a self-supervised deep imputer before any model sees them.

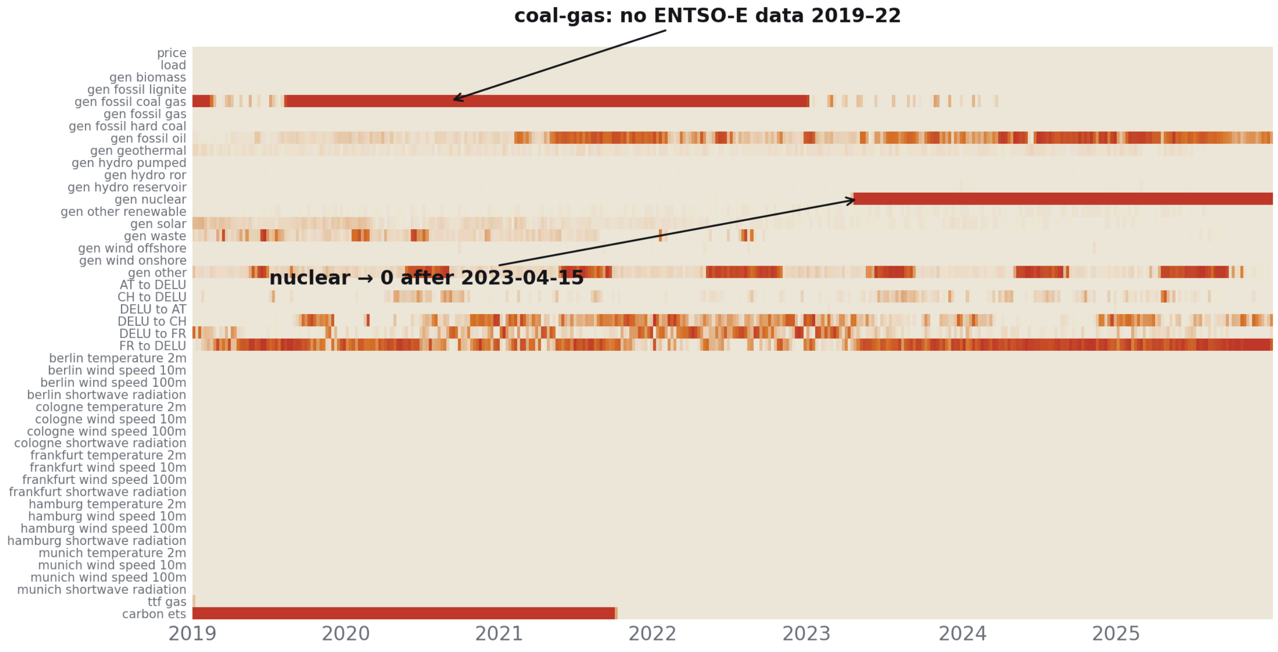
DATA · SOURCES

Six external data sources

SOURCE	SIGNAL	ENDPOINT / NOTE
ENTSO-E	Day-ahead price (€/MWh)	A44 · market zone
ENTSO-E	Actual load (MW)	A65 · physical zone
ENTSO-E	Generation · 17 fuels (MW)	A75
ENTSO-E	Cross-border flows · FR/AT/CH	A11
Open-Meteo	Weather · 5 German cities	temp · wind · radiation
Yahoo / Energy-Charts	TTF gas & EU-ETS carbon	daily fuel prices

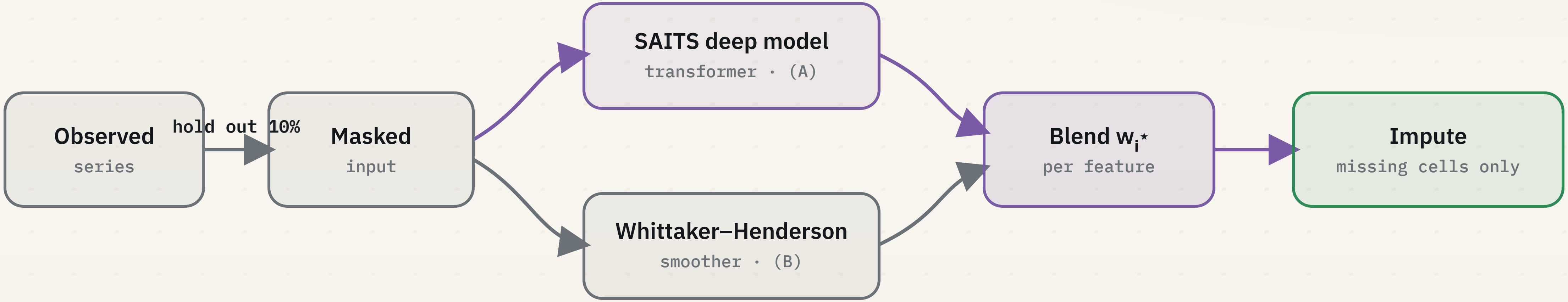
61,368 hourly rows | 2019–2025 | 47 → 52 columns | two EIC codes · market ≠ physical

Missing data is structured, not random



- **Coal-gas:** no ENTSO-E records 2019–2022; the data does not exist.
- **Nuclear → 0**: a real 2023-04-15 shutdown, not a missing value.
- Scattered holiday & exchange gaps: short and recoverable.
- Long structural gaps cannot be recovered; a naïve fill would inject fake signal.

SSHB: self-supervised held-out blend



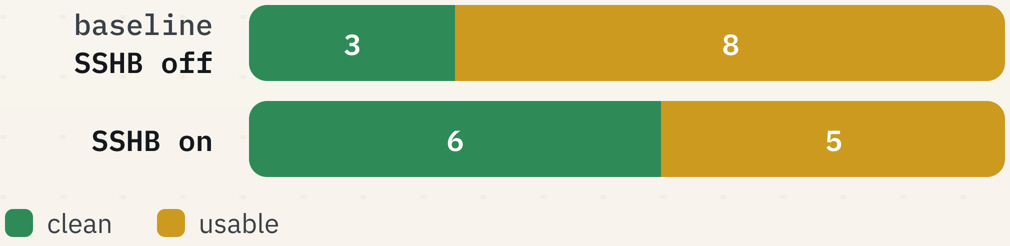
A transformer learns to **reconstruct** held-out cells; a smoother captures local trend. We **blend them per feature**. Where a series is unrecoverable, the weight collapses to zero, and observed values are never overwritten.

More variables made clean, zero regressions

Of 47 columns, only **24 carry real gaps** (generation, flows, fuels); price, load and weather are already complete.

`47 columns` → `24 need imputation` → `11 recoverable` + `13 structural · left untouched`

Audit status · the 11 recoverable variables



+ **13 variables** are structurally unrecoverable: no data exists (coal-gas 2019–22), real shutdowns (nuclear), or variance-collapse (cross-border flows). **Out of scope by design**; SSHB self-limits and leaves them untouched.

3 → 6

recoverable variables **promoted to clean** (+3)

53.9%

of imputed cells **refined** · **0** physical violations

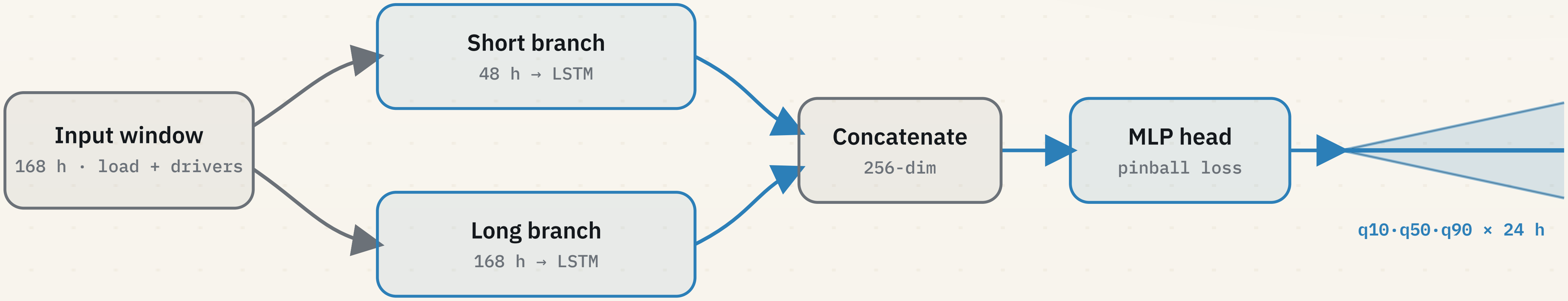
PART 02 · MODULE A

02

Load forecasting

A multi-scale LSTM that emits the first calibrated quantile band of the pipeline.

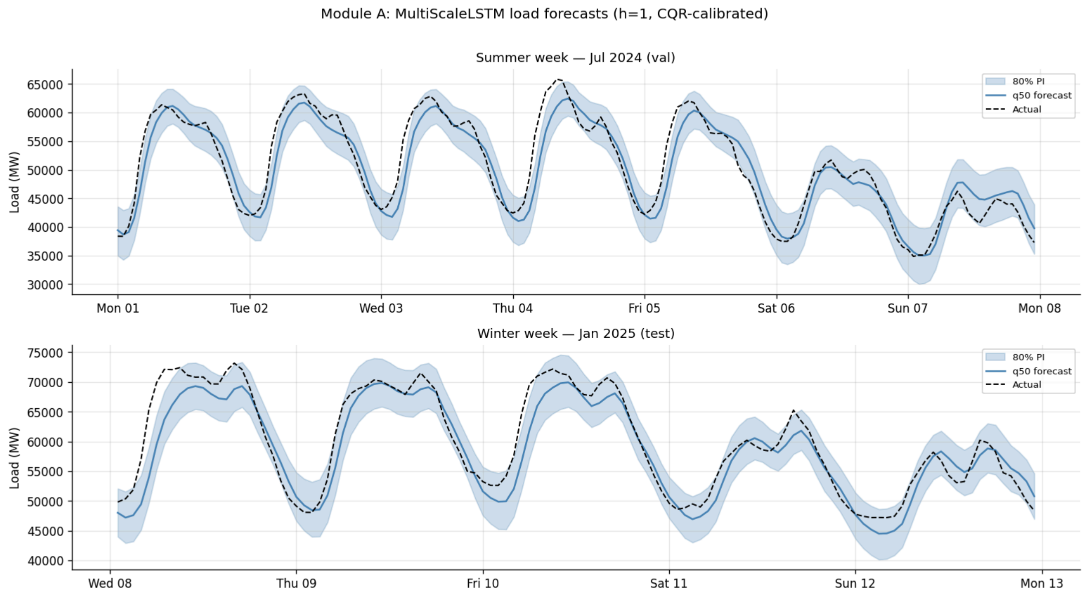
A multi-scale LSTM



Two time-scales capture the daily and weekly load rhythm; CQR then calibrates the band.

MODULE A · RESULTS

Calibrated load forecasts



h+1 forecast band vs. actual load · summer (validation) and winter (test) weeks.

2,591 MW

q50 MAE · ≈ 5% relative error (nMAE)

−5.3%

vs. the 1-week seasonal-naive baseline

54→79%

interval coverage after CQR (δ +1,596 MW)

PART 03 · MODULE B · PRODUCTION

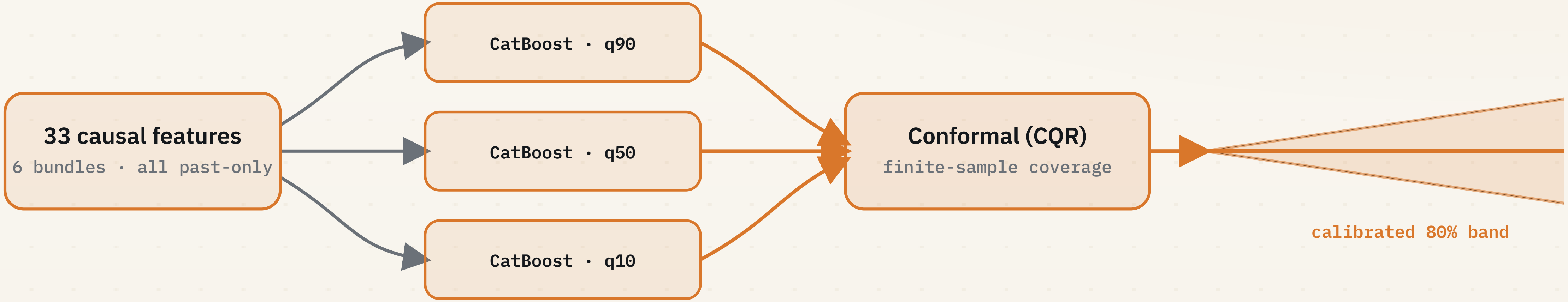
03

Price forecasting

Per-quantile gradient boosting with conformal, finite-sample calibrated intervals.

MODULE B · ARCHITECTURE

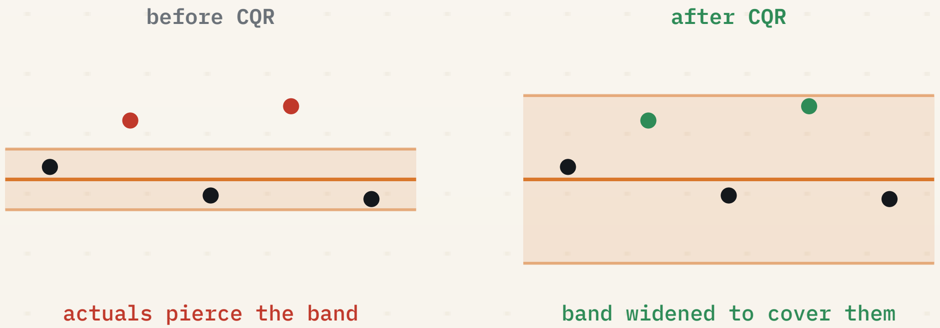
Per-quantile boosting, then conformal



We tried a deep LSTM, but on this repetitive, categorical-heavy data **gradient boosting won**: three independent boosters, one per quantile, then conformal calibration.

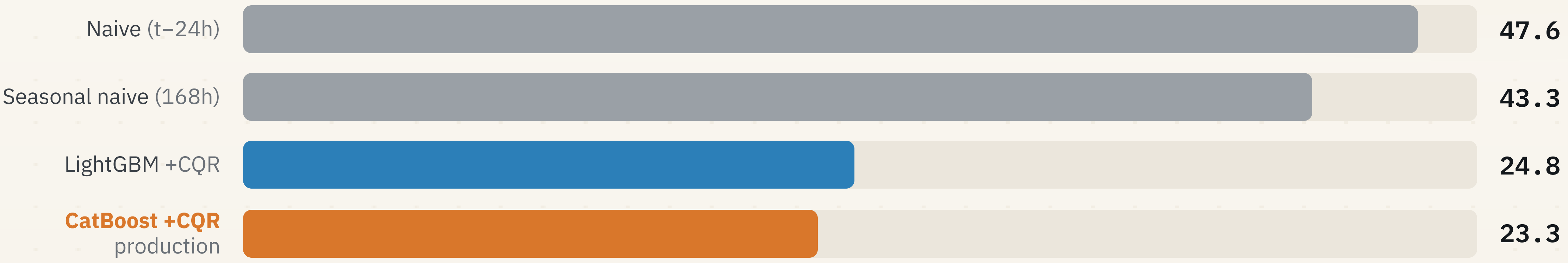
calendar · price lags · spreads · spark / dark · spike flags · 2022 regime · weather

We fixed our own miscalibration



- Raw boosters were **over-confident**: actuals pierced the band.
- Conformal (CQR) widens it with the **⌈(n+1)(1−α)⌉ / n** error quantile, a finite-sample coverage guarantee.
- Upper-tail miss **35.5% → 6.8%** (ideal 10%); interval correction down **54%**.
- We target an **80% band (q10/q90)**, not 95%: the extreme tails are rare spikes, so 80% is the robust interval.

46% better than seasonal-naive



test-set MAE (€/MWh), h1–6 · locked 2025 Q1

MAE 23.29 €/MWh | pinball 7.93 | coverage 0.82 | Diebold–Mariano $p < 0.001$

Load did not help price

≈ 0%

change in price MAE when Module A's load forecasts are added as features

- We wired A into B, measured, and found **no material gain**.
- We report the null result rather than bury it.
- Load *will* help, but later, at the dispatch stage.

● PART 04 · MODULE C

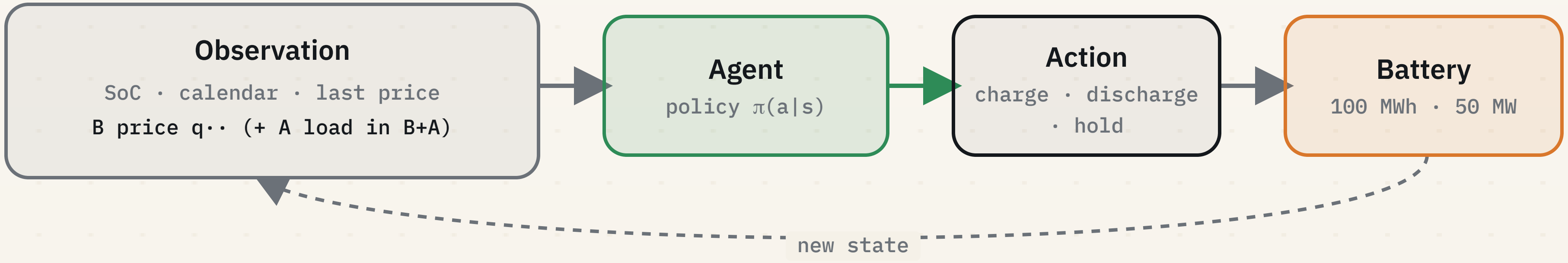
04

Battery dispatch

A risk-aware reinforcement-learning agent trading a battery on the propagated price band.

MODULE C · ARCHITECTURE

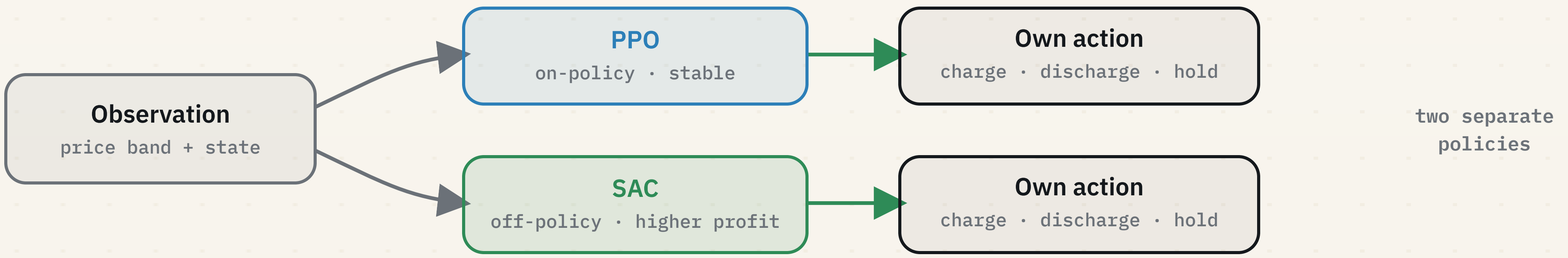
Environment, agent, reward



reward $r_t = p_t \cdot a_t \cdot \Delta t$ (step profit) $- \lambda \cdot \max(0, -\mathrm{CVaR}_{5\%})$, $\lambda = 0.1$

Two agents, trained independently

An agent observes the state, takes an action, and learns from the reward. We train **two** of them and **compare** them; they never interact.



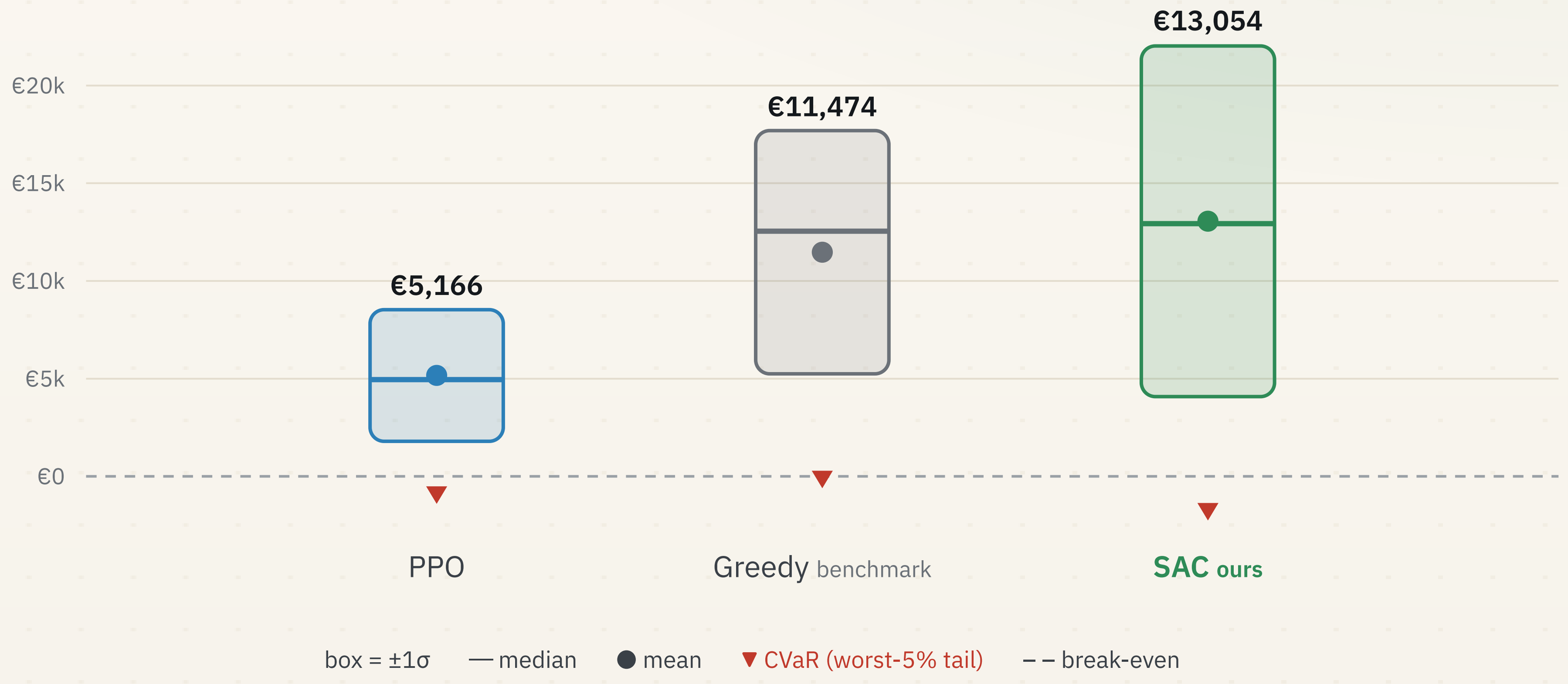
PPO conservative: modest but reliable

SAC aggressive: higher profit, wider swings

200k training steps

78-dim observation

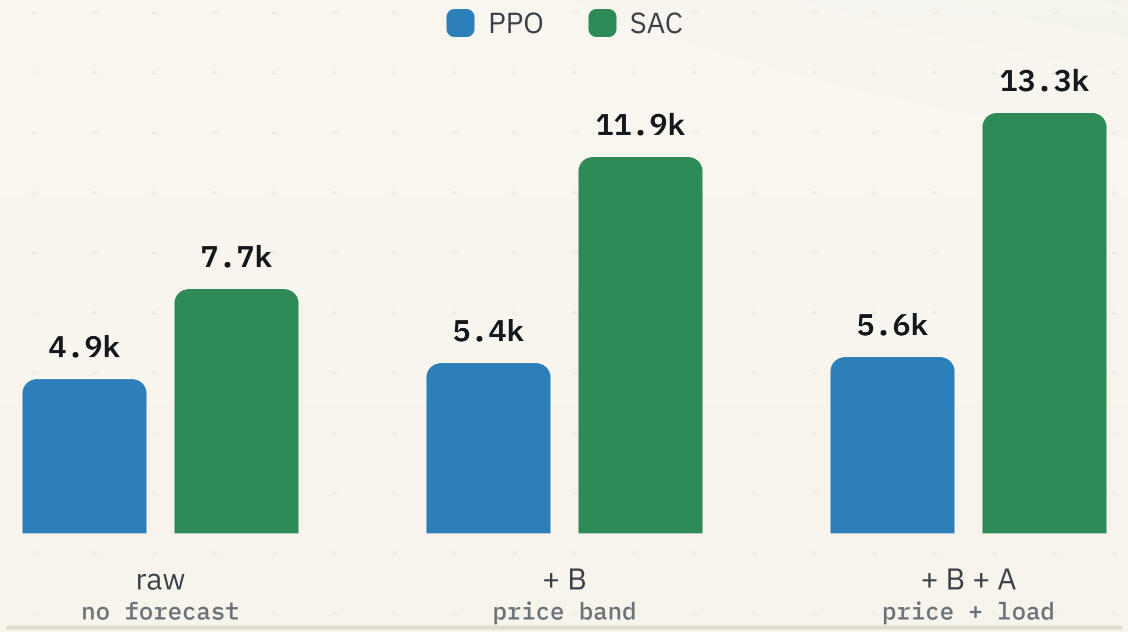
SAC beats the perfect-foresight benchmark



Profit per 24-h episode · 200 test episodes (2025 Q1). **PPO is the most stable; SAC earns the most, above the benchmark on forecasts alone.**

MODULE C · ABLATION

Where each forecast earns its keep



B forecasts → SAC +55%

+ A load → +€1,412 (+12%)

Sharpe 1.43 → 1.72

PART 05

05

Synthesis

What calibrated uncertainty, propagated end to end, is ultimately worth.

CONCLUSION

Uncertainty, propagated, becomes profit

23.29 €/MWh

price MAE · **−46%** vs. seasonal-naive

€13,054

SAC **mean** profit / episode, above the benchmark

53.9%

of imputed cells refined · **0** violations

Calibrated quantiles, carried end to end, turn forecasts into **dispatch profit**.

• ENERGY MARKET INTELLIGENCE SYSTEM

Thank you.

Do you have any questions?

Details & references

SPLIT	PERIOD	HOURS
train	2019–2023	43,824
val	2024	8,784
test	2025 Q1 · locked	2,160

- Imputer: SAITS / Glocal-IB · 4 layers, d=512, 168-h windows.
- Module B: CatBoost depth 8/6/8, ℓ_2 1/3/1; CQR.
- Module C: PPO & SAC (SB3), 78-dim obs, 200k steps; agents trained independently.
- Bands target **80% (q10/q90)**, not 95%: extreme tails are rare spikes.
- Time-series CV with a 24-h gap (no lag-24 leakage).

Methods & data

- ENTSO-E Transparency · Open-Meteo · Yahoo / Energy-Charts.
- SAITS (Du et al., 2023) · Glocal-IB.
- Conformal Quantile Regression (Romano 2019; Barber 2021).
- PPO (Schulman 2017) · SAC (Haarnoja 2018) · CVaR (Rockafellar & Uryasev 2000).
- PyTorch · PyPOTS · CatBoost · scikit-learn · SB3 · Gymnasium.